

Project Sera — Codebase Architecture & Node Representation

This document provides a visual and structural node representation mapping how every Python module and configuration component in the **Project Sera** application connects to one another.

1. Subsystem Architecture Overview

[object Object]

2. Detailed File-to-File Dependency Matrix

Core & Application Controller Layer

[object Object]

UI Shell & Window Routing Layer

[object Object]

Dialog & Extension Interop Layer

[object Object]

3. Component Responsibility Reference

Module Path	Primary Class / Functions	Connected Dependencies	Responsibility
<code>main.py</code>	<code>SeraApp</code> , <code>SyncSignalBridge</code>	<code>database</code> , <code>security</code> , <code>sync_peer</code> , <code>version</code> , <code>app_shell</code> , <code>windows/*</code>	Application entrypoint, vault initialization, Qt signal bridging, auto-updater trigger, window routing.
<code>database.py</code>	<code>SeraDatabase</code> , <code>DatabaseError</code>	<code>security</code>	SQLCipher database CRUD, master column layout (MCL), audit logging, backup/restore, Syncthing/peer conflict matching.
<code>security.py</code>	<code>derive_key_hex</code> , <code>load_salt</code> , <code>verify_pin</code>	Python standard crypto libraries	PBKDF2 key derivation, salt generation/loading, Argon2id PIN verification.
<code>sync_peer.py</code>	<code>SyncPeerService</code> , <code>PeerInfo</code>	<code>main</code>	Zero-configuration UDP LAN peer discovery (<code>BEACON_PORT 49156</code>) & TCP raw database/salt push (<code>SYNC_PORT 49157</code>).
<code>version.py</code>	<code>check_for_updates</code> , <code>apply_and_restart</code>	<code>update_dialog</code>	Queries GitHub raw release metadata (<code>version.json</code>) and orchestrates mandatory application updating.
<code>ui/shell/app_shell.py</code>	<code>AppShell</code>	<code>sidebar</code> , <code>slide_panel</code> , <code>toast</code> , <code>alert_service</code>	Main application shell frame, tab switcher blur effects, notification alert queue.
<code>ui/shell/sidebar.py</code>	<code>Sidebar</code>	<code>theme</code>	Left navigation bar, admin mode toggle, profile row with Sera Sync trigger.

Module Path	Primary Class / Functions	Connected Dependencies	Responsibility
<code>ui/shell/slide_panel.py</code>	<code>SlidePanel</code>	Qt animation framework	Smooth sliding drawer component for viewing client details over search results.
<code>ui/windows/search_window.py</code>	<code>SearchWindow</code>	<code>database</code> , <code>masking</code>	Global client search, quick copy columns, tabular grid, client action shortcuts.
<code>ui/windows/client_detail_window.py</code>	<code>ClientDetailWindow</code>	<code>database</code> , <code>manual_credentials_dialog</code>	Full client profile workspace, credentials, filing history, audit records.
<code>ui/windows/admin_window.py</code>	<code>AdminWindow</code>	<code>database</code> , <code>sera_sync_dialog</code> , <code>mcl_manager_dialog</code> , <code>service_manager_dialog</code> , <code>csv_import_dialog</code>	System administration panel for MCL, CSV operations, backup/restore, and launching Sera Sync.
<code>ui/windows/dashboard_window.py</code>	<code>DashboardWindow</code>	<code>database</code>	DRS (Document Return Status) dashboard grid and compliance status tracking.
<code>ui/dialogs/sera_sync_dialog.py</code>	<code>SeraSyncDialog</code>	<code>sync_peer</code>	Admin modal dialog displaying online LAN peers and executing database push actions.
<code>native_host/host.py</code>	<code>main</code>	Chrome Native Messaging API	Binary STDIN/STDOUT native host bridge forwarding credentials to Chrome/Edge extension.
<code>ui/extension_listener.py</code>	<code>ExtensionListener</code>	<code>database</code>	Local socket server listening for extension auto-fill requests and logging audit actions.
<code>build_tools/build_package.py</code>	Automated script	PyInstaller, spec file, CRX packer	Bundles executable directory (<code>package_dist/Amas_Sera</code>) and packages Chrome extension <code>.crx</code> .
<code>build_tools/installer_setup.iss</code>	Inno Setup script	<code>package_dist/Amas_Sera</code>	Compiles single-file Windows setup installer (<code>Amas_Sera_Setup_vX.X.X.X.exe</code>).

4. Key Execution & Data Flow Patterns

A. Application Initialization & Vault Unlock

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B. LAN Database Synchronization (Sera Sync)

[object Object]

C. Browser Extension Credentials Injection

[object Object]